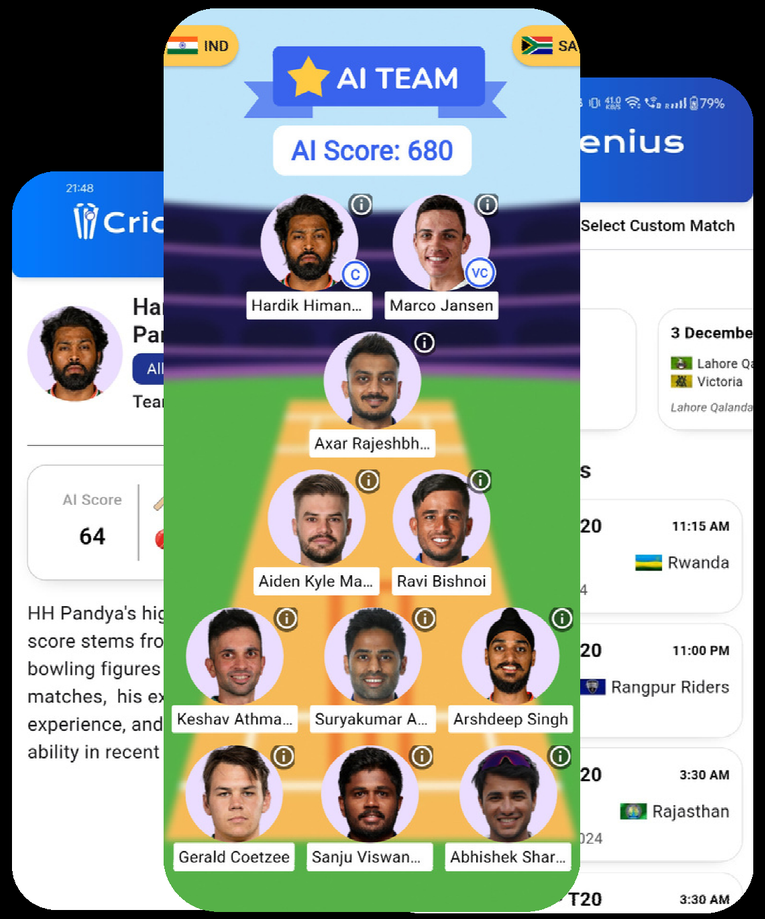


CRICGENIUS

CRICKET TEAM BUILDING WITH PREDICTIVE AI

Presented By

Team 39

OBJECTIVE



Team Prediction

Build a predictive AI Model using historical data along with contexual factors for predicting the performance .

Explainability

Methods/Techniques to explain predictions in a User Friendly way

Interactive UI

Design an intuitive UI for showing AI predicted team and help users in choosing their own team.

OVERVIEW

- Data Processing
- Model Experiments
- Results
- Explainability
- User Interface

DATA PROCESSING

- Cricsheet data
- Cricketstats package
- The format used are as follows:
  - ODI and ODM → OD
  - T20 and IT20 → T20
  - Test and MDM → Test

FEATURES

Batting

- Runs
- Fours
- Sixes
- Dot Balls
- Milestones
- Strike Rate
- etc,

Bowling

- Maiden Overs
- Economy Rate
- LBW/Bowled wickets
- Milestones
- etc,

Fielding

- Direct Runouts
- Indirect Runouts
- Catches
- Stumpings
- etc,

- Venue, Opponent, League specific info
- Venue and League stats (avg. runs, wickets etc.)

FEATURES

- Cumulative Statistics
- Rolling Averages
  - k value ~ 15
- Derived metrics
  - Batting Average
  - Dismissal Rate
  - Dot balls bowled %
- Ray tune

MODEL EXPERIMENTS

- Regressor Models(Random Forest, Spline etc.)
- Deep Learning Models
  - MLP(Multi Layer perceptron)
- Inherently explainable models
  - GAMs (Generative Additive Models)

METRICS

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$MAPE = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

$$\text{Overlap Score} = \frac{1}{n}\sum_{i=1}^{n}\mathbb{I}(y_i = \hat{y}_i)$$

RESULTS

Models comparison

Model	MSE(OD)	MSE(T20)	MSE(Test)
Random Forest	1783.5	1188.8	3933.4
Spline	1679	1065.6	3718.7
MLP	1681.6	1064.3	3712.5
NodeGA2M	1748.7	1157.2	3940.6

MLP results

Format	MAE	MAPE	Overlap Score
OD	380.8	0.38	6.48
T20	272.31	0.30	6.69
Test	538.55	0.29	6.29

EXPLAINABILITY

- Importance of each feature
  - Integrated gradients
  - Gradient SHAP
- Top 5 important features
- User-friendly explanations ~ LLM (Gemini 1.5 flash)
- Voice Output

EXPLAINABILITY PROMPT

You are a cricket fantasy prediction explainability assistant. Your task is to explain the fantasy score prediction based on the most important features.
.....
Example of positive and negative explainable texts can be: *Player's high predicted fantasy score is attributed to their feature1, feature2, and feature3.* and *Player's strong performance in feature1, feature2, and feature3 was commendable, but competition was tough, leading to their exclusion.*
Just give the result no starting text.

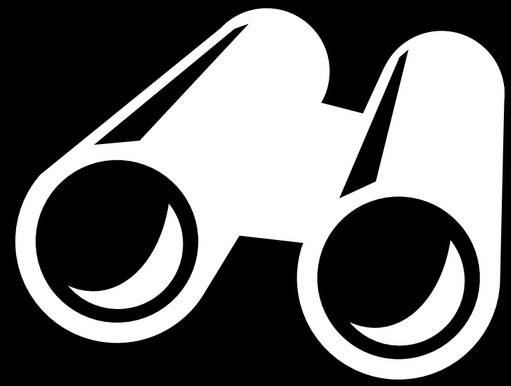
{*Prediction Explanation for Fantasy Score*

Top contributing features:

HH Pandya
Selected
1. *Feature:* last_15_matches_Balls Bowled_sum
- *Type:* Main
- *Importance Score:* 10.665008814752122
2. *Feature:* Total_matches_played_sum
- *Type:* Main
- *Importance Score:* 8.005089060766462
..... }

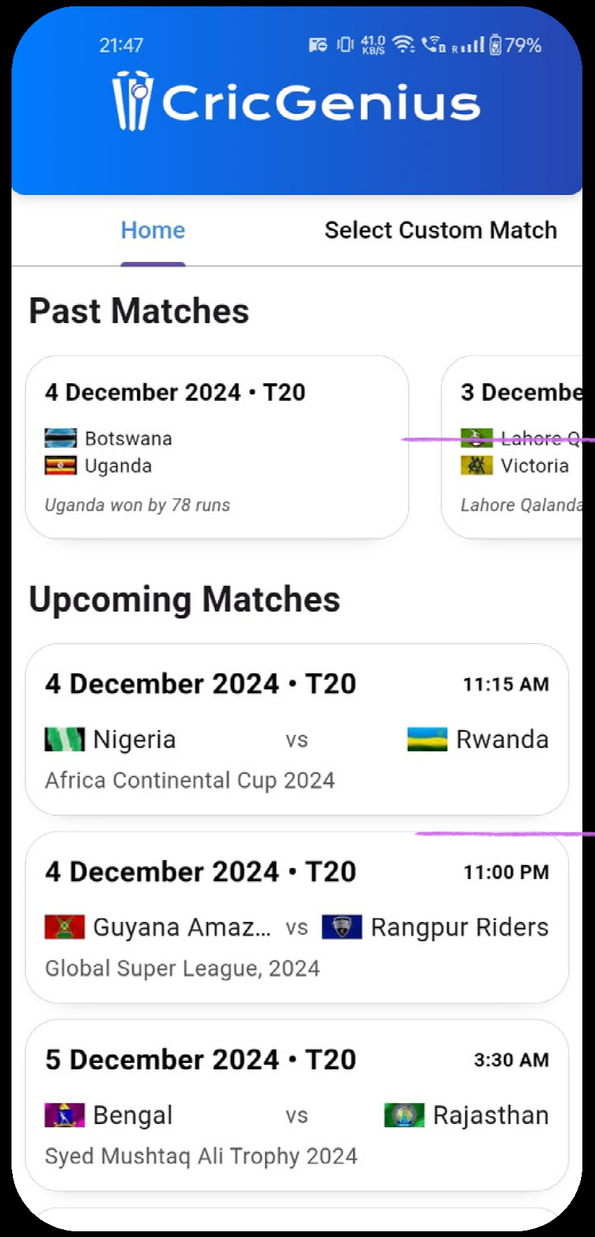
Give priority to the features with lower index since they have a bigger score, and output the individual text explanation for each player, separated by a separator "
.....

FUTURE SCOPE



- Tuning the feature to get best possible feature set which are important for good prediction
- The model can be trained exclusively for all 6 formats.

PRODUCT UI
Home Screen



Team Names with images

Match venue details

Match Results

List of Past Matches
Along with the final result

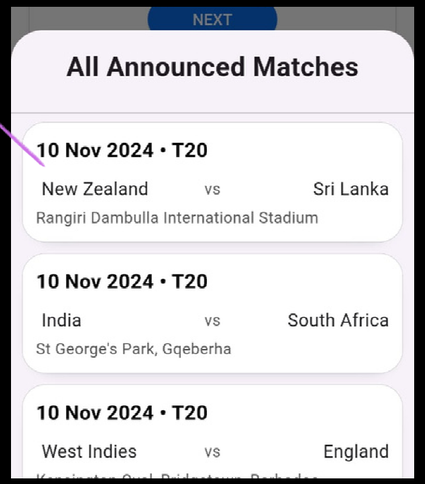
List of all upcoming matches for upto 4 days

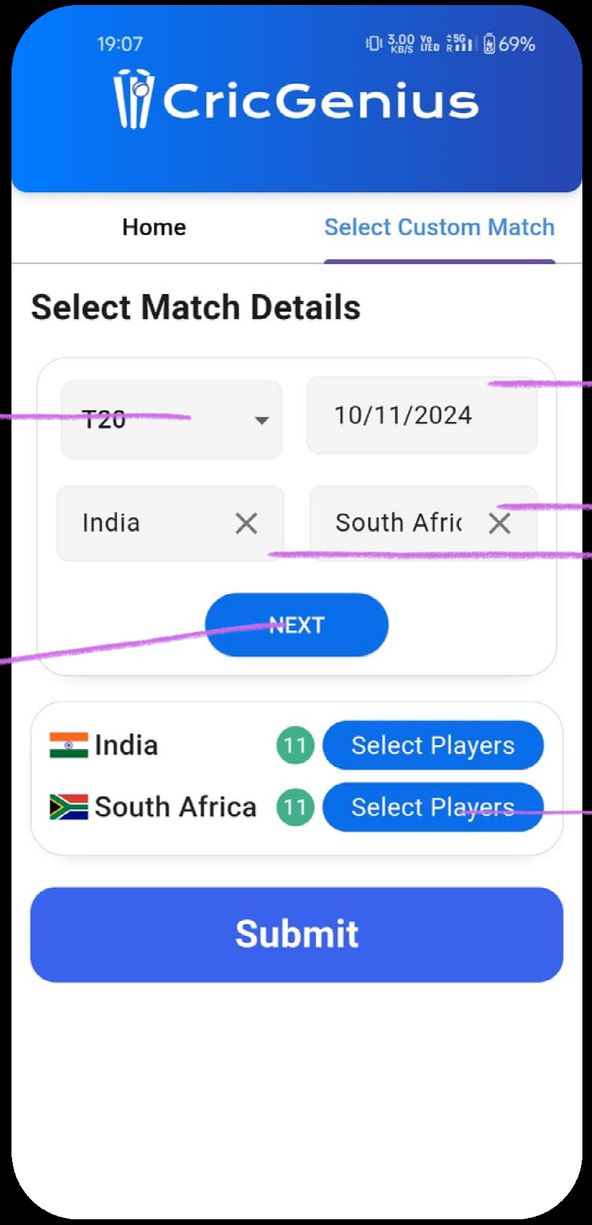
this can directly take users to the custom match selection screen to show the predicted AI Team

PRODUCT UI
Match Selection Screen

Match Format

Clicking next button shows the list of announced matches with that format and date





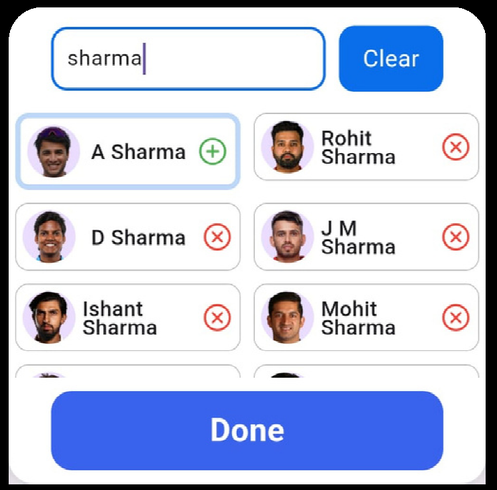
Match Date

Select Team Names. Autocomplete is also provided for filling easily

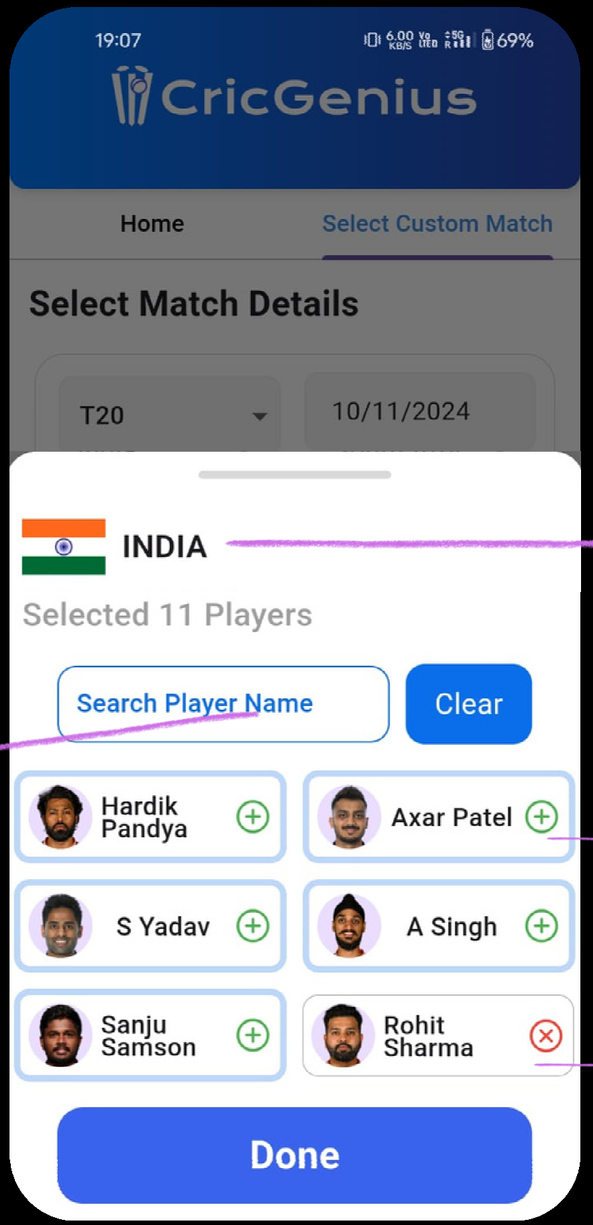
Choose the list of 11 players for the model to predict Dream Team

PRODUCT UI
Player Selection Modal

Player Search in action



Player Search



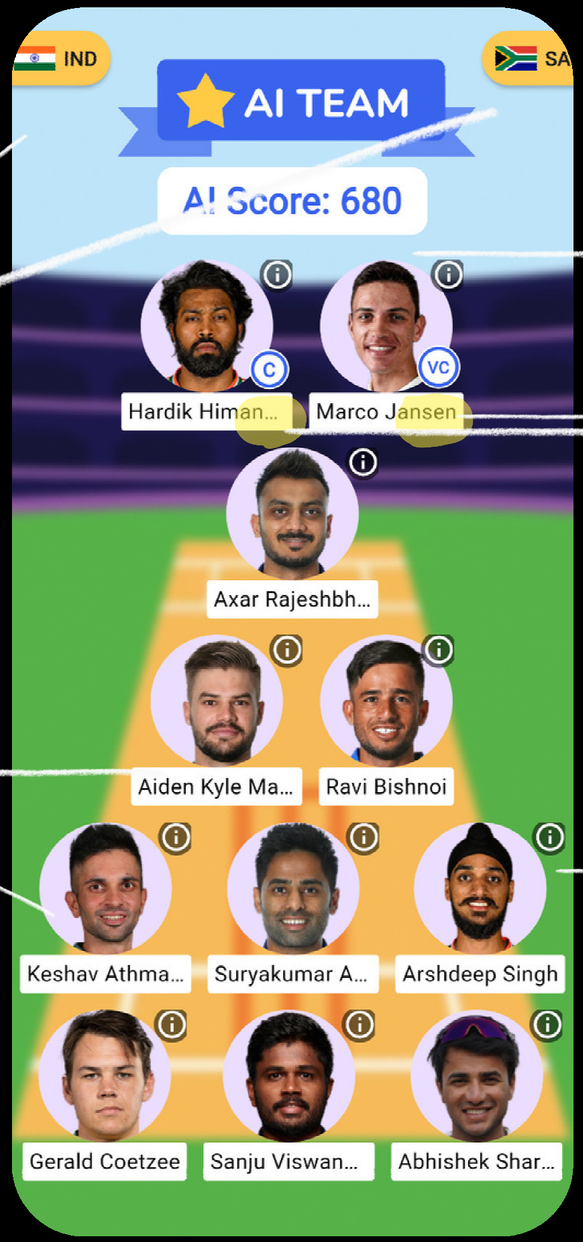
Team Details

Selected Player

Excluded Player

PRODUCT UI
AI Team Screen

Team Details are shown in an intuitive way



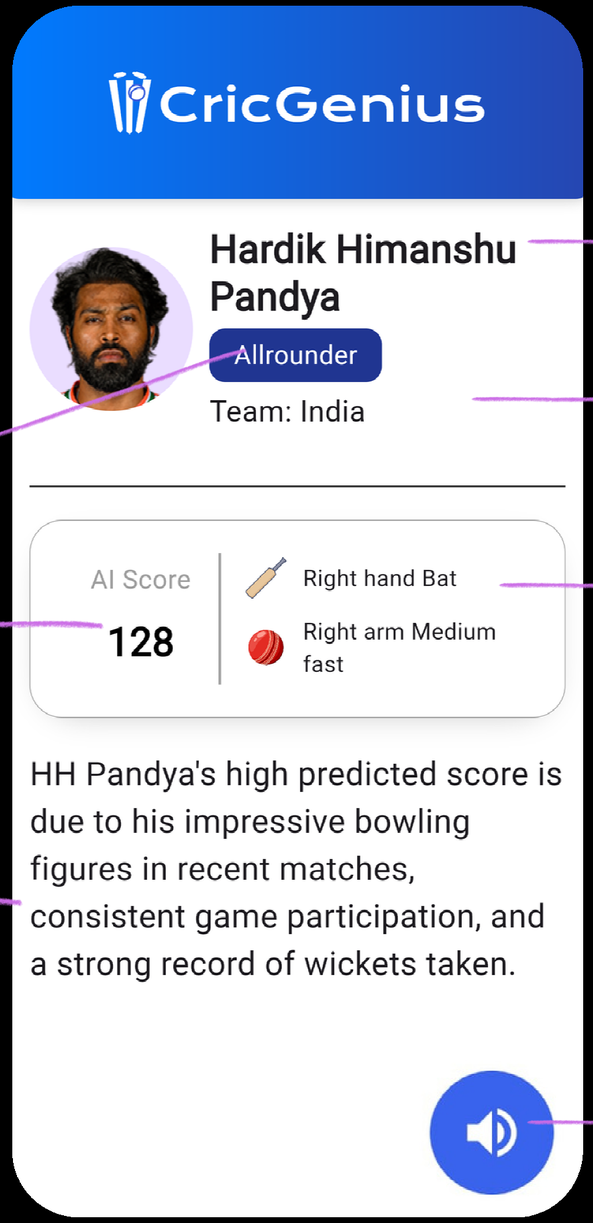
Total AI Predicted Fantasy Points

Recommended captain and vice captain are shown in the same page.

Player photos are shown to users for easy identification

Info button is added which takes to the screen with more details about the player selection

PRODUCT UI
Player Info Screen



Player Name

Team Name

Player Role

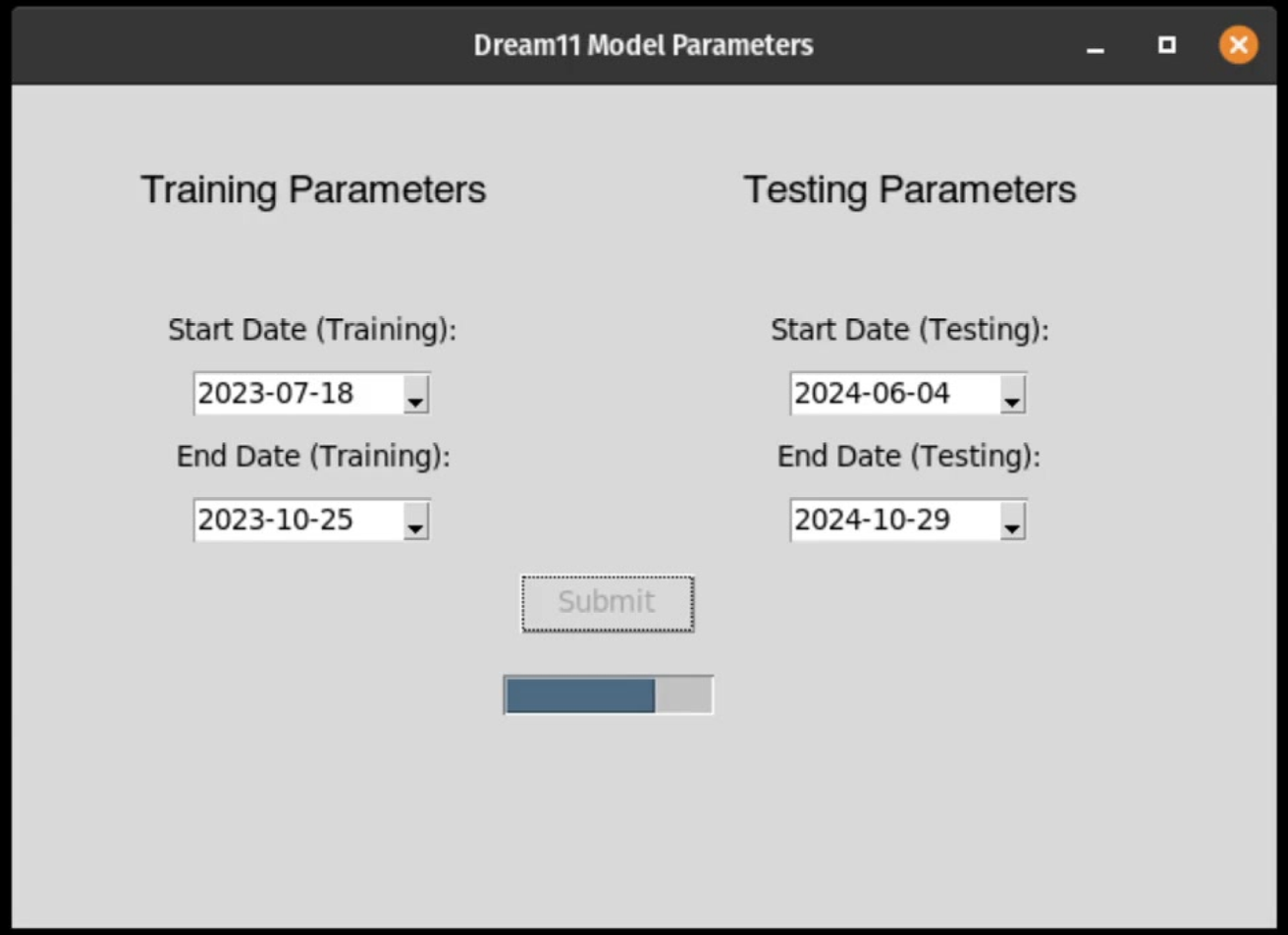
Info about players batting and bowling style

AI predicted fantasy points

Explanation for why the player was selected

Audio based explanation. Can be extended to use different language based on user preference.

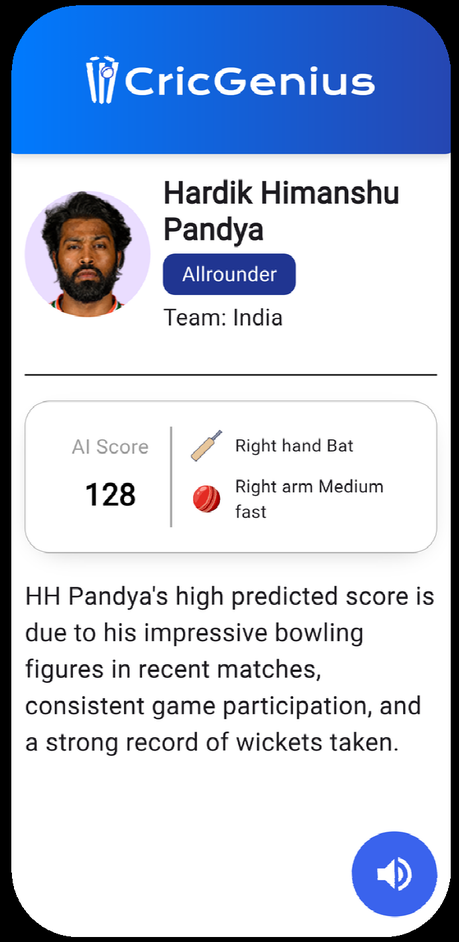
MODEL UI


Dream11 Model Parameters
Training Parameters
Testing Parameters
Start Date (Training):
2023-07-18
End Date (Training):
2023-10-25
Start Date (Testing):
2024-06-04
End Date (Testing):
2024-10-29
Submit

PRODUCT UI
Scope for Improvements



Option for users to make their own team and show the comparison to the predicted Team



Different Language can be shown to serve users all over India

More stats with graphics can be shown to better explain the AI selection

THANK YOU!!